

AgiBot X2 Ground Recovery

Simulation, reinforcement learning and ROS 2 integration | 29 September 2026

Result: 5/5 supine recoveries. The selected PPO policy was trained from random weights in one uninterrupted 1,500-update run. Every reset is supine. All five 10-second validation episodes end in stable two-foot stance without other-body support. No upright-start pretraining or lift assistance entered this run.

Robot and simulation

Item	Implementation
Model	Official AgiBot X2 Ultra v1.3.0 URDF, upstream revision 60c5de5; locally converted to USD.
Physics	Isaac Sim 6.0.1 / Isaac Lab 3.0.0-beta2.patch1 / PhysX; floating base, 200 x 200 m flat floor, self-collision.
Constraints	31 movable joints and 32 sensed bodies; vendor meshes, inertia, joint axes and actuator limits. Position commands use imported 98% soft limits.
Reset / rate	Supine pelvis height 0.190 m, neutral joints and zero velocity; collision clearance checked. Physics 200 Hz, policy 50 Hz; 10 s or 500 policy steps.

Fetch and import: `scripts/fetch_agibot_model.sh` and `scripts/import_x2_isaac.sh`. The generated USD stays local. Mesh collision shapes use the converter's convex-hull approximation.

Observation and action contract

Policy input	Count
Pelvis height; body-frame linear/angular velocity; projected gravity	1 + 3 + 3 + 3
Joint position / velocity	31 + 31
Left/right foot contact; whole-body ground-contact bits	2 + 32
Previous two policy actions	16
Total	122

Eight symmetric channels command hip pitch, knee, ankle pitch, shoulder pitch, elbow, waist pitch, ankle roll and hip roll. They vary 15 joint targets using $\text{centre} + \text{span} * \tanh(\text{action})$, then clamp to imported soft limits. Separate normalized actor/critic MLPs use 512/256/128 ELU layers.

The other 16 joints retain default targets and remain simulated and published: hip yaw; shoulder roll/yaw; wrist roll/pitch/yaw (both sides); waist roll/yaw; head pitch/yaw. Actions specify positions, not direct torques.

Evaluation seeds

One checkpoint, five reproducible supine starts. Each seed samples X/Y within +/-4 cm, Z within +/-1 mm, roll/pitch within +/-0.005 rad and yaw within +/-0.02 rad. Joints remain neutral and velocity zero. The table shows measured differences at reset.

Seed	Initial pelvis height	Initial projected gravity Z	End strict stance
101	0.1896 m	-0.0017	8.96 s
102	0.1905 m	+0.0012	8.94 s
103	0.1894 m	+0.0001	8.96 s
104	0.1905 m	+0.0046	8.88 s
105	0.1902 m	-0.0006	8.86 s

Reward design and PPO experiment

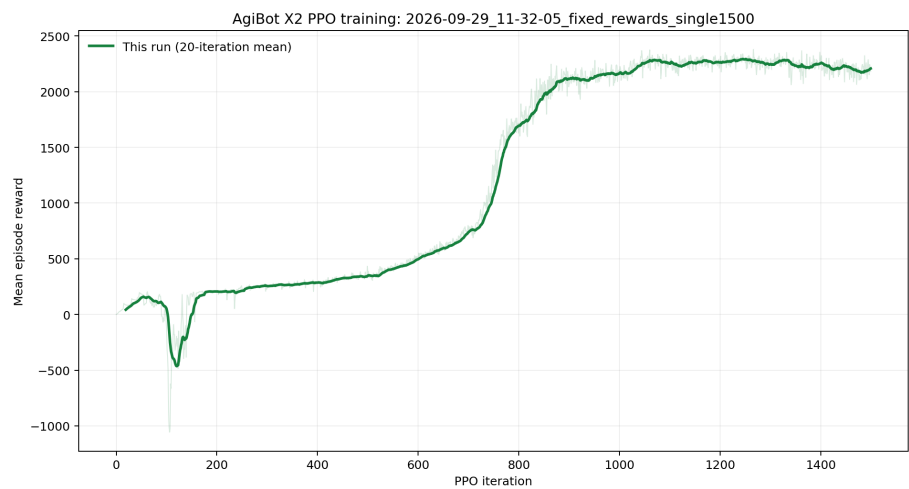
Signed height guides the rise without rewarding an inverted bridge. Transitional hand/body contacts remain allowed. Dense posture and motion/command regularization guide the transition from rising to sustained foot-only stance. These terms and weights were fixed for all 1,500 updates.

Term	Weight	Purpose / implementation
Signed pelvis height	40	$\text{clip}(z/0.68) * \text{clip}((1-g_z)/2)$.
Head height / upright	5 / 5	Clipped exponential head height; $\exp(-g_z)$.
Feet / other support	10 / -1	Two-foot contact; discourage other support near standing.
Balance / strict stance	20 / 20	Low velocity near standing; complete physical success predicate.
Stance proximity	20	Exponential proximity to supported stance. Training widths: tilt 0.35, linear speed 0.8 m/s, angular speed 2 rad/s.
Leg pose	20	Neutral hip pitch/knee/ankle pitch; gated above 0.40 m and by upright orientation.
Relaxed arms	40	Supported-upright shoulder/elbow pose near 0 / -0.15 rad; variance 2.
Shoulder command	80	Supported shoulder raw action near $\text{atanh}(0.15)$, variance 1.
Near-stance motion	-2	Penalize squared base speed and 0.1 times squared angular speed near standing.
Command saturation	-0.5	Squared excess outside effective raw-action limits; accounts for tanh and imported soft-limit clipping.
Action change / joint speed	-0.02 / -0.0005	Squared successive raw-action changes / joint velocities.
Torque / limit / failure	-1e-6 / -1 / -10	Squared effort, soft-limit violation, non-timeout safety termination.

RewardManager sums weight x term x 0.02 s. HumanUP [1] motivates rise/control terms; HoST [2] motivates post-task posture; FRASA [3] motivates compact symmetric control. Weights, gates, widths and command targets are local X2 choices, not paper constants.

Training configuration

PPO [4]: clip 0.2, gamma 0.99, GAE lambda 0.95; 5 epochs, 4 minibatches, value coefficient 1, gradient clip 1, desired KL 0.01. One process trains 3,000 environments for 1,500 updates with 32 steps/environment, seed 47, initial action std 0.8 and adaptive learning rate. Random actor/critic weights, no parent checkpoint and fixed reward terms throughout.



Complete single-run reward history: iterations 0-1499, faint logged values and dark 20-update mean. The independent physical evaluation, not reward magnitude, establishes recovery.

Five-episode validation

A recovery requires every condition below for at least 0.5 continuous seconds. Evaluation continues for the complete episode; terminal samples are captured before automatic reset. Final stance duration is reported separately from a brief earlier success.

Check	Threshold
Height	Pelvis at least 0.58 m.
Orientation	$\text{norm}(\text{projected_gravity_xy}) \leq 0.15$ and $g_z \leq -0.98$.
Base motion	Linear speed ≤ 0.25 m/s; angular speed ≤ 0.35 rad/s.
Support	Each foot ≥ 15 N ground force; every other body < 15 N.

Measured outcome

Seed	Recovery	Strict stance at end	Two feet at height peak
101	Yes	8.96 s	No
102	Yes	8.94 s	No
103	Yes	8.96 s	No
104	Yes	8.88 s	No
105	Yes	8.86 s	No

5/5 successful recoveries. All five end on both feet without other-body support, with 8.86-8.96 s of uninterrupted strict stance at episode end. Both feet briefly leave the floor at the height peak, then land for sustained unsupported stance. This transient is a limitation of the controller; it is excluded from strict-standing duration.

Source: `reports/relaxed_v4_evaluation.json`. Checkpoint: `reports/checkpoints/x2_supine_single1500.pt`.

Failure analysis and scope

An early height-focused supine experiment scored 0/5 at 500 updates and 0/5 in a separate 2,000-update test: rising or bouncing did not become stable foot-only stance. A mixed-start prototype succeeded but is excluded from the submitted policy. These comparisons do not isolate any single causal change.

A dense-stance continuation reached 5/5 after 1,000 inherited updates. To give the submitted checkpoint, graph and video a simple shared provenance, the final fixed reward preset was retrained from random weights in one 1,500-update run. This is a new independent experiment, not a resume of the older policy.

The selected run uses a local, supported shoulder/arm preference from update 0. It does not use the historical forward-arm continuation. No imitation dataset, lift force, reset curriculum or upright-start pretraining enters this checkpoint.

Scope: nominal flat-floor simulation with small reset perturbations. Terrain variation, broad pose recovery, sensor noise, latency and real-hardware performance were not validated. Development failures and exact configuration provenance remain in Git and `docs/development_history.md`.

ROS 2 interface and reproduction

The Python package builds with colcon and launches recovery and telemetry nodes together. A local Unix socket connects ROS 2 Humble (Python 3.10) to the Isaac policy server (Python 3.12). Joint positions come from the live simulator.

Interface	ROS 2 type / behavior
/x2/start_recovery	std_srvs/srv/Trigger: acceptance before execution; a second request while busy returns false.
/x2/recovery_status	std_msgs/msg/String: IDLE, RUNNING, SUCCEEDED or FAILED.
/x2/joint_states	sensor_msgs/msg/JointState: timestamps and 31 measured joint names/positions while running.

Telemetry logs status and left_knee_joint at 1 Hz. ROS ends after the physical 0.5 s success hold; the separate full-length evaluation verifies final stance. Timeout is configurable. A fresh isolated colcon build passed; the selected exported policy was tested through the real ROS-Isaac connection.

Recorded check	Observed outcome
CLI recovery	RUNNING to SUCCEEDED with timestamped live 31-joint telemetry.
Concurrent request	First accepted; second returned false: Recovery already running.
Timeout 0.2 s	RUNNING to FAILED after 10 steps with timeout 0.2 s.

Commands from the repository root

Install the pinned dependencies using README.md and docs/commands.md. Training is optional for supplied-checkpoint playback.

```
export ISAAC_PYTHON=/path/to/isaac/environment/bin/python
./scripts/fetch_agibot_model.sh
./scripts/import_x2_isaac.sh
CKPT=reports/checkpoints/x2_supine_single1500.pt
./scripts/play_isaac.sh "$CKPT" --environment relaxed_v4 \
  --device cuda:0 --viz kit --start-delay 2 --seeds 101 \
  --output /tmp/x2_play.json
./scripts/evaluate_isaac.sh "$CKPT" --environment relaxed_v4 \
  --device cuda:0 --output /tmp/x2_evaluation.json
```

The two-second preview freezes the initial pose; physics and the episode clock do not advance. Seeds select small pose perturbations. Each new training run saves its checkpoint, reward PNG/CSV and manifest under its own timestamped logs/rsl_rl directory.

For ROS: build with scripts/build_ros.sh, start the Isaac policy server, launch both ROS nodes in a separately prepared terminal, then call /x2/start_recovery. Ordered commands and expected outputs are in docs/commands.md and docs/ros_live_validation.md.

Evidence and references

Evidence: reports/ (weights, training graph, evaluation and ROS logs); docs/parameter_provenance.md (parameter sources); docs/development_history.md (failures and fixes). Tested host: Ubuntu 22.04.5, RTX 5080 Laptop GPU with 16 GB.

[1] HumanUP, RSS 2025. doi:10.15607/RSS.2025.XXI.063

[2] HoST, RSS 2025. doi:10.15607/RSS.2025.XXI.064

[3] FRASA, arXiv:2410.08655, Section IV-D

[4] Proximal Policy Optimization, arXiv:1707.06347